Canyu Zhang

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EDUCATION

NYU Shanghai, Shanghai, China **May 2027**

B.S. in Computer Science | GPA: 3.93/4.0 | Major GPA: 3.98/4.0

- ***Graduate-Level Courses:*** Deep Learning, Computer Vision, NLP, Efficient AI System, Compiler Construction
- ***Mathematics for CS:*** Linear Algebra, Multivariable Calculus, ODE, Discrete Mathematics, Probability & Statistics

RESEARCH EXPERIENCE

InfoFlow KV: Information-Flow-Aware KV Recomputation for Long Context **Jun. 2025 - Feb. 2026**

**Co-first author*/ICML 2026. (with Prof. Shengjie Wang, Prof. Tianyi Zhou, and Prof. Danyang Zhuo)*

- Built a **chunk-based** KV prefilling and recomputation pipeline enabling reusable KV caches and recovering cross-chunk attention under long-context inference across multiple LLMs (Qwen, LLaMA, ChatGLM).
- Proposed two information-flow-aware recomputation strategies: a **query-conditioned attention-norm** token selection method aligned with inference-consistent RoPE geometry, and a **chunk reordering** scheme that improves prompt–context interaction, achieving up to **+6%** F1 score on HotpotQA over prior methods.
- Analyzed how RoPE positional geometry affects token selection in chunk-wise long-context inference, identifying frequency range and prompt–context proximity as key factors and evaluating four representative RoPE configs.

Generative Visual Memory for Long Egocentric Videos **Mar. 2026 - Present**

Research Assistant/ Ongoing(with Prof. Mengye Ren)

- Exploring **generative memory**, which stores a long egocentric video in a video generation model's weights and regenerates relevant visual evidence from a textual cue when the original video is unavailable.

Submodular Context Partitioning and Compression for In-Context Learning **Mar. 2025 - Aug. 2025**

Research Assistant/ (with Prof. Shengjie Wang, and Prof. Tianyi Zhou)

- Studied **Sub-CP**, a **block-aware context selection framework** for in-context learning that leverages submodular objectives to **balance global diversity and local coherence** across demonstration context blocks.
- Implemented **ensemble-based** evaluations under the **DENSE ICL framework** using Mistral-7B, benchmarking Sub-CP against uniform and random partition strategies across **five standard NLP datasets**.
- Achieved up to **+7.1%** average F1 score over baseline partition methods and **+15.4%** F1 score on TREC.

SAGE: Benchmarking and Improving Retrieval for Deep Research Agents **Sep. 2025 - Feb. 2026**

Research Assistant/EACL 2027, under review. (with Prof. Chen Zhao)

- Built **SAGE**, a benchmark for reasoning-intensive scientific literature retrieval with **1,200 queries across four domains and a 200K-paper corpus**, supporting **short-form and open-ended questions**.
- Evaluated six deep research agents under both **web search and corpus retrieval settings**, revealing that BM25 outperforms LLM-based retrievers by **~30%** due to keyword-oriented sub-query generation.
- Proposed a **corpus-level test-time scaling framework** that augments documents with metadata and keywords using LLMs, improving retrieval accuracy by **+8% (short-form)** and **+2% (open-ended)**.

TECHNICAL SKILLS

Programming: Python, C/C++, OCaml | **Languages:** English, Mandarin

Frameworks & Tools: PyTorch, HuggingFace Transformers, vLLM, FlashAttention, FlashInfer, Git, Linux